



Finals – Section 3 Information Pack

Section 3: Reward It

30 Marks available in this Section – Available time is 30 minutes

Everyone's raw mark for this section will be scaled up by an equal proportion such that the top scorer is awarded 100% of the 30 marks.

INTRODUCTION

This section requires you to build a model to calculate Base Fees and Performance Fees on behalf of a Fund Manager.

ModelOff Funds Management ("MFM") invests money in different long-term assets on behalf of contributors to the fund. For managing the fund, MFM is entitled every six months to both a Base Fee and, if certain hurdles are met, a Performance Fee. The Fund was initiated on 1 January 2011, and data is provided up to and including 31 December 2015. The Fund's base currency is British Pounds (GBP).

Questions 1 – 4 relate to the Base Fee and Questions 5 – 9 relate to the Performance Fee. The calculations for the Base Fee and the Performance Fee are essentially independent, so completing the Base Fee is not required to attempt the Performance Fee questions (and vice versa).

TASK

Your task is to build a model that can be used to calculate all historical fee payments (up to and including 31 December 2015) that MFM were due based on the data given. A workbook is provided containing some of the assumption data. The remaining assumption data is provided in this written document.

COMMON INFORMATION

The information below is required for both the Base Fee and Performance Fee calculations.

Exchange Rates

You have been provided exchange rates between all relevant currencies and the British Pound since the inception of the fund. Exchange rates are given monthly and apply for the entire calendar month specified.

Asset Purchases

Assets are purchased at certain times as chosen by MFM. Each asset will have a designated local currency, which may be different to the fund's base currency of GBP. The asset purchase price will be used as the Asset Valuation value (see below) from the purchase date until the next quarter end date.

You have been provided the dates, amounts and currencies of Asset Purchases since the inception of the fund.

Asset Valuations

On the final day of every quarter, all assets in the Fund are valued in units of the local asset currency. These valuations will apply until the following quarter end when a new valuation is made.

You have been provided with local currency valuations of each of the fund's assets since the inception of the fund.



Finals – Section 3 Information Pack

BASE FEE

[Base Fee – Definitions](#)

Uncalled Committed Capital

Investors commit capital to the fund at various times. When capital is committed it is designated as “Uncalled Committed Capital”. When MFM wishes to make an investment (asset purchase) it draws on the pool of Uncalled Committed Capital.

You have been provided the dates and amounts of capital commitments, beginning with the inception of the fund on 1 January 2011.

Fund Net Asset Value:

On any given date, the Fund Net Asset Value is the combined sum of all Asset Valuations expressed in GBP, calculated using the exchange rate that applies on that date. This includes all relevant Asset Purchases that have not yet had a quarter-end Asset Valuation undertaken.

[Base Fee – Calculation Methodology](#)

Beginning on 1 January 2011, MFM is entitled to a base fee that is calculated daily and paid twice a year (every 30 June and 31 December). The base fee is calculated as 1.00% per annum of Uncalled Committed Capital plus 1.50% per annum of Fund Net Asset Value. The accrued fee should be calculated daily (using the end of day values that take into account any capital commitments, asset purchases, asset valuations or exchange rate changes that occur on a given day). The outstanding accrued amount is then paid at the end of the day every 30 June and 31 December. Apply the per annum % rates on an actual/365 basis for daily pro-rata purposes.

PERFORMANCE FEE

[Performance Fee – Definitions](#)

Performance Fee Calculation Date

Each 30 June and 31 December that falls after 1 January 2011.

Hurdle Value

On each Performance Fee Calculation Date, the Hurdle Value equals the sum over all assets purchased of:

$$A * (1 + \text{Hurdle Rate})^{(n/365)}$$

Where:

A = Asset Purchase Price expressed in GBP using the historical exchange rate that applied at the date of purchase.

Hurdle Rate = 7.00%.

n = the number of days between the Asset Purchase Date and the Performance Fee Calculation Date (calculated as the Performance Fee Calculation Date minus the Asset Purchase Date).

Hurdle Return

On each Performance Fee Calculation Date:

Hurdle Return equals: the Hurdle Value minus Invested Capital.



Finals – Section 3 Information Pack

Total Return

On each Performance Fee Calculation Date:

Total Return equals: Adjusted Fund Net Asset Value minus Invested Capital.

Adjusted Fund Net Asset Value:

On each Performance Fee Calculation Date, Adjusted Fund Net Asset Value is the combined sum of all Asset Valuations expressed in GBP, calculated using the historical exchange rate that was in force at the date of acquisition of the asset.

Invested Capital

On each Performance Fee Calculation Date, the combined sum of the purchase price of Assets in the fund, expressed in GBP using the historical exchange rate that applied at the date of purchase.

Performance Fee – Calculation Methodology

On each Performance Fee Calculation Date, the Performance Fee Earned and the Performance Fee Deficit are calculated as follows:

- If Total Return $\leq$ Hurdle Return, then:
 - Performance Fee Earned = 0; AND
 - Performance Fee Deficit = Hurdle Return minus Total Return
- If 1.25 times Hurdle Return $\geq$ Total Return $>$ Hurdle Return, then:
 - Performance Fee Earned = Total Return minus Hurdle Return; AND
 - Performance Fee Deficit = 0
- If Total Return $>$ 1.25 times Hurdle Return, then:
 - Performance Fee Earned = 20% * Total Return; AND
 - Performance Fee Deficit = 0.

Performance Fee – Notional Account

On each Performance Fee Calculation Date:

- The Performance Fee Earned from the period is added to the balance of the notional Performance Fee Account.
- The Performance Fee Deficit from the period is subtracted from the balance of the notional Performance Fee Account.
- After the above movements have happened, any balance above zero in the notional Performance Fee Account is paid to MFM as a Performance Fee Paid.

The notional Performance Fee Account begins at a balance of zero on 1 Jan 2011. The balance of this account can become negative.



MODELOFF

TM

Financial Modeling World Championships 2015

Finals – Section 3 Information Pack

Your Name: _____

QUESTIONS – BASE FEE

Question 1 (3 Marks)

To the nearest pound, what is the balance in GBP of the Uncalled Committed Capital on 10 May 2011?
The answer will be between 2,300,000 and 2,400,000.

Answer:

Model reference (sheet and cell/s):

Question 2 (3 Marks)

To the nearest pound, what is the total Fund Net Asset Value (in GBP) on 31 July 2012?
The answer will end with the digits 41.

Answer:

Model reference (sheet and cell/s):

Question 3 (4 Marks)

To the nearest pound, what is the Base Fee (in GBP) paid on 30 June 2011?
The answer will be 5 digits, and the digits will sum to 23.

Answer:

Model reference (sheet and cell/s):

Question 4 (4 Marks)

To the nearest pound, what is the total Base Fee paid (in GBP) in calendar 2014?

Answer:

Model reference (sheet and cell/s):



MODELOFF

TM

Financial Modeling World Championships 2015

Finals – Section 3 Information Pack

Your Name: _____

QUESTIONS – PERFORMANCE FEE

Question 5 (2 Marks)

To the nearest pound, what is the Hurdle Value (in GBP) calculated on 31 Dec 2013?
The answer will be between 9,600,000 and 9,700,000.

Answer:

Model reference (sheet and cell/s):

Question 6 (2 Marks)

To the nearest pound, what is the Total Return (in GBP) calculated on 30 Jun 2015?
The answer will end in the digits 74.

Answer:

Model reference (sheet and cell/s):

Question 7 (4 Marks)

To the nearest pound, what is the Performance Fee Deficit (in GBP) calculated on 31 Dec 2011?
The answer will be between 19,000 and 20,000.

Answer:

Model reference (sheet and cell/s):

Question 8 (4 Marks)

To the nearest pound, what is the Performance Fee Earned (in GBP) calculated on 31 Dec 2013?

Answer:

Model reference (sheet and cell/s):

Question 9 (4 Marks)

To the nearest pound, what is the total Performance Fee Paid (in GBP) over the 5 years from 1 Jan 2011 to 31 Dec 2015 (in GBP)?

Answer:

Model reference (sheet and cell/s):
